Intervals	1 Ionian	2 Dorian	3 Phrygian	4 Lydian	5 Mixolydian	6 Aeolian	7 Locrian
24	$\mathbf{I}s2$	$\mathbf{ii}s2$	—	$\mathbf{IV}s2$	$\mathbf{V}s2$	$\mathbf{vi}s2$	—
33	$\mathbf{I}$	$\mathbf{ii}$	$\mathbf{iii}$	$\mathbf{IV}$	$\mathbf{V}$	$\mathbf{vi}$	$\mathbf{vii}^{\circ}$
34	$\mathbf{vi}^{1}$	$\mathbf{vii}^{\circ 1}$	$\mathbf{I}^{1}$	$\mathbf{ii}^{1}$	$\mathbf{iii}^{1}$	$\mathbf{IV}^{1}$	$\mathbf{V}^{1}$
42	$\mathbf{I}s4$	$\mathbf{ii}s4$	$\mathbf{iii}s4$	—	$\mathbf{V}s4$	$\mathbf{vi}s4$	—
43	$\mathbf{IV}^{2}$	$\mathbf{V}^{2}$	$\mathbf{vi}^{2}$	$\mathbf{vii}^{\circ 2}$	$\mathbf{I}^{2}$	$\mathbf{ii}^{2}$	$\mathbf{iii}^{2}$
44	—	$\mathbf{ii}q$	$\mathbf{iii}q$	—	$\mathbf{V}q$	$\mathbf{vi}q$	$\mathbf{vii}q$
233	$\mathbf{ii}7^{3}$	$\mathbf{iii}7^{3}$	$\mathbf{IV}\Delta^{3}$	$\mathbf{V}7^{3}$	$\mathbf{vi}7^{3}$	$\mathbf{vii}^{\varnothing}7^{3}$	$\mathbf{I}\Delta^{3}$
323	$\mathbf{IV}\Delta^{2}$	$\mathbf{V}7^{2}$	$\mathbf{vi}7^{2}$	$\mathbf{vii}^{\varnothing}7^{2}$	$\mathbf{I}\Delta^{2}$	$\mathbf{ii}7^{2}$	$\mathbf{iii}7^{2}$
332	$\mathbf{I}6$	$\mathbf{ii}6$	$\mathbf{I}\Delta^{1}$	$\mathbf{IV}6$	$\mathbf{V}6$	$\mathbf{IV}\Delta^{1}$	$\mathbf{V}7^{1}$
333	$\mathbf{I}\Delta$	$\mathbf{ii}7$	$\mathbf{iii}7$	$\mathbf{IV}\Delta$	$\mathbf{V}7$	$\mathbf{vi}7$	$\mathbf{vii}^{\varnothing}7$
334	$\mathbf{I}^{+}8$	$\mathbf{ii}^{+}8$	$\mathbf{iii}^{+}8$	$\mathbf{IV}^{+}8$	$\mathbf{V}^{+}8$	$\mathbf{vi}^{+}8$	$\mathbf{vii}^{+}8$
433	$\mathbf{IV}^{2+}8$	$\mathbf{V}^{2+}8$	—	—	$\mathbf{I}^{2+}8$	—	—
434	$\mathbf{ii}7^{3}$	$\mathbf{iii}7^{3}$	$\mathbf{IV}\Delta^{3}$	$\mathbf{V}7^{3}$	$\mathbf{vi}7^{3}$	$\mathbf{vii}^{\varnothing}7^{3}$	$\mathbf{I}\Delta^{3}$
444	—	$\mathbf{ii}q7$	$\mathbf{iii}q7$	—	—	$\mathbf{vi}q7$	$\mathbf{vii}q7$

*Chord figure = column degree + row intervals (e.g., col **5** × row 333 → figure 5333 = $\mathbf{V}7$).*

Key

Upper = major (I,IV,V)
Lower = minor (ii,iii,vi)
Δ = major 7th
$^{+}8$ = root doubled at 8ve
$^{\circ}$ = diminished
$^{\varnothing}7$ = half-dim
7 = dominant 7th
$s4^{+}8$ = sus4 w/ doubled root
q = quartal
$s2/s4$ = sus2/sus4
$6/m6$ = major/minor 6th
^{1,2,3} = inversion

Creating the Chord Figure

333 / 5333 = *intervals* / *deg+intervals*
5333 = deg 5, three thirds = $\mathbf{V}7$
1233 = deg 1, 2nd+2 thirds = $\mathbf{I}\Delta^{1}$
1433 = deg 1, 4th+2 thirds = $\mathbf{I}s4^{+}8$

How to read. Rows are voicing shapes (generic intervals bottom-up: 2=2nd, 3=3rd, 4=4th). Columns are diatonic degrees 1–7. A cell is the Roman-numeral chord that results (case = major/minor; suffix = quality). A *figure* concatenates column-degree + row-intervals (5333 = $\mathbf{V}7$). Superscripts 1/2/3 mark inversion (bass chord tone). — = non-diatonic. **To voice a chord** ($\mathbf{G}7$ in $\mathbf{C}$): express as Roman ($\mathbf{V}7$), find the cell, stack the row intervals from the bass note.

ii / I	Warm IΔ9	133 933
iii / I	Soft IΔ	133 A33
V / I	Bright I6/9	133 533
vi / I	Pastoral I6	133 633
IV / I	Mixolydian blur	133 B33
IΔ / I	Full maj7 stack	133 8333
vii$^\circ$ / I	Tense IΔ	133 733
I$s4$ / I	Sus4 hang	133 842
I$s2$ / I	Sus2 open	133 824
I / IΔ	Doubled tonic	1333 833
IV / IΔ	Lydian tonic	1333 B33
iii / IΔ	Sparse IΔ9	1333 A33
ii / IΔ	Full jazz tonic	1333 933
V / IΔ	Gospel tonic	1333 C33
V / ii	ii7+11	233 C33
IV / ii	ii7+9	233 B33
I / ii	ii7	233 833
vi / ii	Rich	233 633
iii / ii	Mournful	233 A33
I / IV	IVΔ9	433 833
V / IV	Lydian	433 C33
ii / IV	IV add 6/9	433 933
IΔ / IV	Double-Δ	433 8333
iii / IV	IVΔ9 (alt)	433 A33
vi / IV	IVΔ warm	433 D33
IV$s2$ / IV	Sus2 IV	433 B24
V / IVΔ	IVΔ9$\sharp$11	4333 C33
ii / IVΔ	IVΔ9+6	4333 G33
IV / V	V11 modal	533 B33
ii / V	V7+9	533 933
vii$^\circ$ / V	V7 bite	533 E33
vi / V	Warm V7+9	533 D33
I / V	V$s4$ hang	533 F33
iii / V	V add 6	533 A33
V$s4$ / V	Sus4 V	533 C42
V$s2$ / V	Sus2 V	533 C24
Vq / V	Quartal V	533 C44
V / V7	V7 thick	5333 C33
IV / V7	V11 suspended	5333 B33
vi / V7	V13 alt	5333 D33
ii / V7	Jazz V79	5333 G33
I / V7	V13	5333 F33
ii7 / V7	Jazz ii–V	5333 G333
IΔ / V7	Cadence approach	5333 F333
IVΔ / V7	Jazzy pre-cadence	5333 B333
iii / vi	vi7+9	633 A33
I / vi	vi7 pad	633 F33
IV / vi	vi$m$6	633 B33
V / vi	Aeolian modal	633 C33
ii / vi	Spacious minor	633 G33
vii$^\circ$ / vi	vi$m$(maj7)	633 E33
V / iii	Phrygian tint	333 C33
I / iii	Mellow mediant	333 833
vi / iii	Phrygian add	333 D33
IV / iii	Dark Phrygian	333 B33
V / ii7	Sec. dom of ii	2333 C33
V / vi7	Sec. dom of vi	6333 C33
V7 / IΔ	Prep. cadence	1333 C333
V / IVΔ	Full Lydian	4333 C33
vii$^\circ$ / iii	iii7$\flat$5	333 733
IV / vii$^\circ$	vii$^\varnothing$11	733 B33
vii$^\circ$ / IΔ	IΔ9$\sharp$11	1333 733
I / ii7	iim11 — Dm11 voicing	2333 833
ii / iii$m$7	iiim11	3333 933
iii / IVΔ	IVΔ9$\sharp$11 alt	4333 A33
V / vi7	vim13	6333 C33
vi / vii$^\varnothing$7	vii$^\varnothing$13	7333 D33
I^{+8} / I	Bell-ring I	133 8334
ii^{+8} / ii	Minor bell	233 9334
iii^{+8} / iii	Mediant bell	333 A334
IV^{+8} / IV	Subdominant bell	433 B334
V^{+8} / V	Dominant bell	533 C334
vi^{+8} / vi	Rel.-minor bell	633 D334
vii^{+8} / vii	Leading-tone bell	733 E334

Cycle of 2nds

I ↔ ii	Cloudy	*Grounding*
ii ↔ iii	Deepening	*Easing*
iii ↔ IV	Warming	*Cooling*
IV ↔ V	Tensing	*Calming*
V ↔ vi	Deceptive	*Pivoting*
vi ↔ vii°	Darkening	*Reposing*
vii° ↔ I	Releasing	*Fracturing*

Cycle of 3rds

I ↔ iii	Drifting	*Homecoming*
iii ↔ V	Galvanizing	*Retreating*
V ↔ vii°	Fraying	*Steadying*
vii° ↔ ii	Tilting	*Curdling*
ii ↔ IV	Gladdening	*Shading*
IV ↔ vi	Grieving	*Consoling*
vi ↔ I	Triumphing	*Brooding*

Cycle of 4ths

I ↔ IV	Exploring	*Landing*
IV ↔ vii°	Alarming	*Quieting*
vii° ↔ iii	Despairing	*Rooting*
iii ↔ vi	Accepting	*Reverting*
vi ↔ ii	Wandering	*Dreaming*
ii ↔ V	Lifting	*Stepping*
V ↔ I	Resolving	*Lofting*

The Feedback Loop — every rep

Plan a specific target (not "play it better" — "keep the LH thumb on the string until the RH starts"), **play** honestly at thinking-tempo, **evaluate** strictly (90%=success, 50%=info), **adjust** one variable before the next rep.

The 90/10 Principle

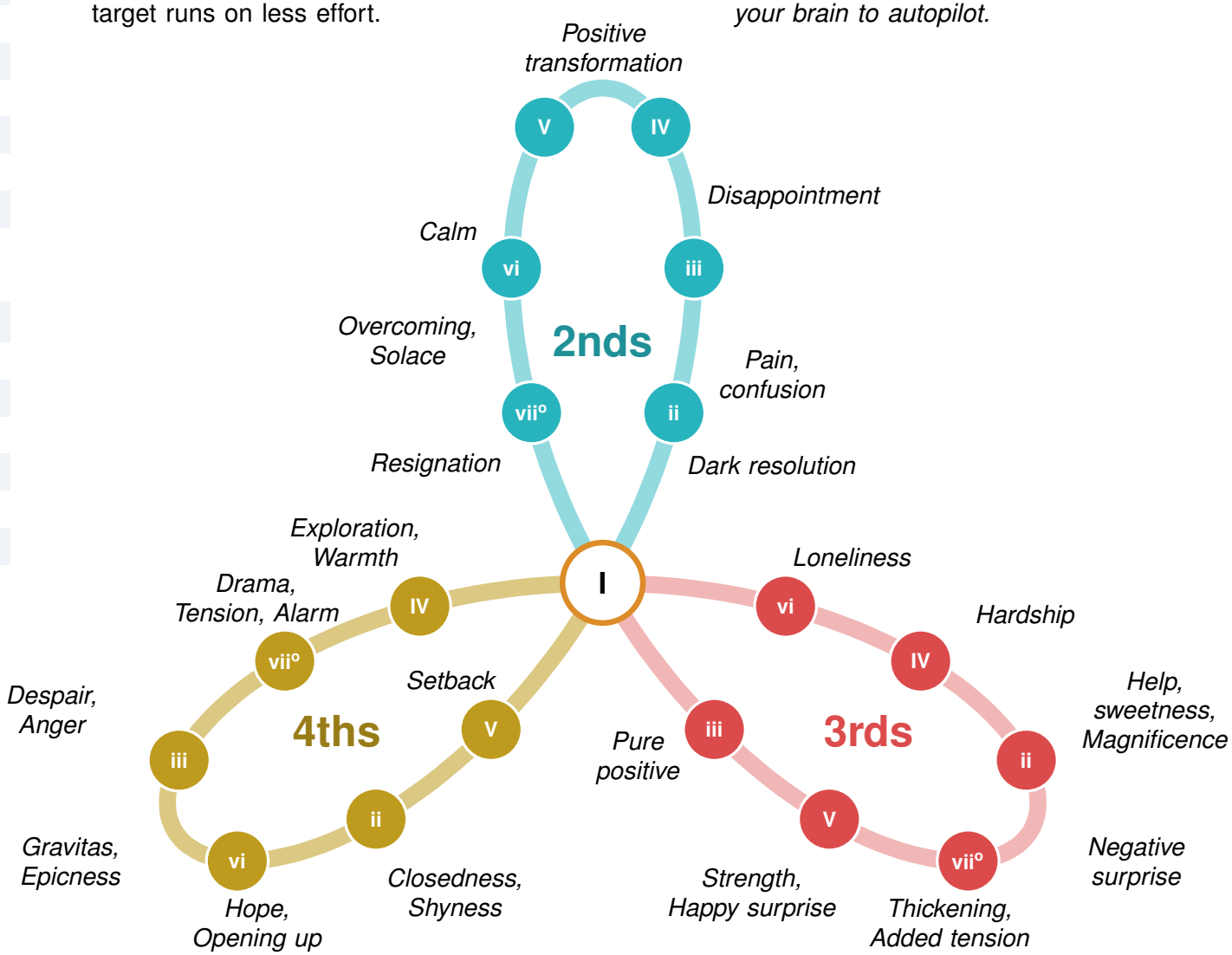
To nail a skill at 100%, push a *harder related skill* to 90%: speak finger numbers while arpeggiating, practice the cascade in reverse, do crescendos on a flat passage. When you drop the extra layer the target runs on less effort.

Five Axes of Difficulty

Raise one axis, lower another, cycle: **Length** (bar→phrase→piece) · **Complexity** (one hand→two→add cascade) · **Depth** (phrasing, dynamics, tone) · **Speed** (last, and the most overused) · **Mental endurance**.

3-4-2-1 Session Order

Rank L1 (automatic) → L4 (impossible). Go **L3→L4→L2→L1**: L3 warm-up demands focus; L4 while sharpest; L2 as attention fades; L1 last for cruise-mode review. *Never start with L1 — it sets your brain to autopilot.*



Cycle 2nd	LH	RH	Figure	CW	CCW
	I	ii	133 933	Cloudy	*Grounding*
	ii	iii	233 A33	Deepening	*Easing*
	iii	IV	333 B33	Warming	*Cooling*
	IV	V	433 C33	Tensing	*Calming*
	V	vi	533 D33	Deceptive	*Pivoting*
	vi	vii°	633 E33	Darkening	*Reposing*
	vii°	I	733 F33	Releasing	*Fracturing*
	I	ii7	133 9333	Cloudy	*Grounding*
	ii*s*2	iii	224 A33	Deepening	*Easing*
	iii*s*4	IV	342 B33	Warming	*Cooling*
	IV	V7	433 C333	Tensing	*Calming*
	V+8	vi	5334 D33	Deceptive	*Pivoting*
	vi	vii∅7	633 E333	Darkening	*Reposing*
	vii°	I+8	733 F334	Releasing	*Fracturing*

Cycle 3rd	LH	RH	Figure	CW	CCW
	I	iii1	133 743	Drifting	*Homecoming*
	iii	V	333 C33	Galvanizing	*Retreating*
	V	vii°	533 E33	Fraying	*Steadying*
	vii°	ii	733 G33	Tilting	*Curdling*
	ii	IV	233 B33	Gladdening	*Shading*
	IV	vi	433 D33	Grieving	*Consoling*
	vi	I	633 F33	Triumphing	*Brooding*
	I6	iii7	1332 A333	Drifting	*Homecoming*
	iii2	V	534 C33	Galvanizing	*Retreating*
	V*q*	vii°	544 E33	Fraying	*Steadying*
	vii∅71	ii	6233 G33	Tilting	*Curdling*
	ii*q*7	IV	2444 B33	Gladdening	*Shading*
	IV1	vi	143 D33	Grieving	*Consoling*
	vi72	I	3323 F33	Triumphing	*Brooding*

Cycle 4th	LH	RH	Figure	CW	CCW
	I	IV	133 B33	Exploring	*Landing*
	IV	vii°	433 E33	Alarming	*Quieting*
	vii°	iii	733 H33	Despairing	*Rooting*
	iii	vi	333 D33	Accepting	*Reverting*
	vi	ii	633 G33	Wandering	*Dreaming*
	ii	V	233 C33	Lifting	*Stepping*
	V	I	533 F33	Resolving	*Lofting*
	I	IV∆	133 B333	Exploring	*Landing*
	IV	vii∅7	433 E333	Alarming	*Quieting*
	vii+8	iii	7334 H33	Despairing	*Rooting*
	iii*s*4	vi	342 D33	Accepting	*Reverting*
	vi7	ii	6333 G33	Wandering	*Dreaming*
	ii7	V	2333 C33	Lifting	*Stepping*
	V6	I	5332 F33	Resolving	*Lofting*